

Personal Notes on Partition Theory

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1 My Propositions

Proposition 1. The number of partitions of n in which positive distinct parts $\lambda_1, \lambda_2, \dots, \lambda_m$ each appear at least once equals the number of partitions of $n - \sum_{i=1}^m \lambda_i$.

Proposition 2 is directly inspired by Chapter 1 examples 1 and 2 from *The Theory of Partitions* by George E. Andrews.

Proposition 2. The following are equal: the number of partitions of n in which multiples of x are distinct, here denoted $p_\alpha(n)$; the number of partitions of n in which no part is divisible by $2x$, here denoted $p_\beta(n)$; the number of partitions of n in which no part appears more than $2x - 1$ times, here denoted $p_\gamma(n)$.

Proof.

$$\begin{aligned} \sum_{n \geq 0} p_\alpha(n) q^n &= \prod_{n \geq 1} (1 + q^{xn}) \frac{1 - q^{xn}}{1 - q^n} \\ &= \frac{(q^{2x}; q^{2x})_\infty}{(q; q)_\infty} \\ &= \sum_{n \geq 0} p_\beta(n) q^n. \end{aligned}$$

By Glashier's Theorem (Theorem 1) $p_\beta(n) = p_\gamma(n)$. ■

2 Theorems

Theorem 1 (Glashier's Theorem). The number of partitions of n in which no part appears more than d times, here denoted $p_\alpha(n)$, equals the number of partitions of n in which no part is divisible by $d+1$, here denoted $p_\beta(n)$.

Proof.

$$\begin{aligned}
 \sum_{n \geq 0} p_\alpha(n) q^n &= \prod_{n \geq 1} (1 + q^n + \cdots + q^{dn}) \\
 &= \prod_{n \geq 1} \frac{1 - q^{(d+1)n}}{1 - q^n} \\
 &= \frac{(1 - q^{d+1})(1 - q^{2(d+1)}) \cdots}{(1 - q) \cdots (1 - q^{d+1}) \cdots (1 - q^{2(d+1)}) \cdots} \\
 &= \prod_{n \geq 1, d+1 \nmid n} \frac{1}{1 - q^n} \\
 &= \sum_{n \geq 0} p_\beta(n) q^n.
 \end{aligned}$$

■